Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/17/2024
PI Name:	Foster, Jamie Ph.D.		
Project Title:	Impact of Spaceflight on Beneficial Animal-Microbe Interactions		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	EDITOR'S NOTE: Congressional district is for University of Florida, Gainesville. Address is PI's physical address.		
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	11/01/2018	**End Date:**	10/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/31/2023 per NSSC information (Ed., 9/12/22) NOTE: End date changed to 10/31/2022 per NSSC information (Ed., 9/15/21)		
Key Personnel Changes/Previous PI:	Eric Koch, who was the postdoctoral fellow, has taken a job and has now left the project; however, he is continuing to help analyze the data. Joseph Emhof is a new Ph.D. student who has joined my laboratory and will be using this data as part of his dissertational research		
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0138		
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Task Description:	Beneficial microbes play a significant role in the normal development of animal tissues. Through the formation of healthy, stable associations between host and symbiont, microbes can induce a wide range of immunological, physiological, and developmental pathways. In this experimental research plan, we are examining the impact of spaceflight on the molecular interactions between host and symbiont that lead to normal bacteria-induced animal development. To address this goal, we are using the established model system between the bobtail squid Euprymna scolopes and the luminescent bacterium Vibrio fischeri. The bacterium colonizes a specialized tissue, referred to as the light organ, enabling the host to use the luminescence in an anti-predator behavior. For more than a quarter-century, this squid-vibrio system has served as an important model to examine the cell-cell interactions between mutualistic bacteria and their animal hosts. Our previous ground-based studies have identified several bacteria-induced developmental phenotypes under modeled microgravity conditions; however, the mechanisms underlying these phenotypes have yet to be fully elucidated. Our previous results also determined that the presence of the mutualistic V. fischeri modulates the host immune response of the squid by down regulating several stress and immune genes. Here, in this study, we propose to replicate and expand upon the previous ground-based studies by partnering with Techshot and using their fluid processing cassette hardware for the manipulation of both the animal and symbiotic partner in situ during spaceflight. Specifically, we propose the following once the spaceflight is complete: Science Objective 1. Characterize the changes in the transcriptome of the host squid in the presence and absence of symbiotic bacteria exposure under microgravity conditions. We propose to generate transcriptomes to host E. scolopes light organs in the presence and absence of the mutualistic V. fischeri at several key developmental time points (0, 2, 6, 12 h) post-colonization. The transcriptomes of symbiotic and aposymbiotic animals exposed to spaceflight at 0 and 12 h will be compared to previously generated transcriptomes (i.e., ground-based) to ascertain whether transcriptional differences are observed between modeled and actual microgravity conditions. Science Objective 2. Ascertain changes in the metabolome of host squid light organ in the presence and absence of symbiotic bacteria under spaceflight conditions. Using the same flight samples as described in Objective 1, an untargeted metabolomics approach will be used to reveal differential metabolic activity in symbiotic and aposymbiotic light organs. Science Objective 3. Build integrative networks of transcriptomes and metabolomes under different flight and ground-based treatments to elucidate crucial pathways that differentially respond to the spaceflight environment. The networks generated from the results of Objectives 1 and 2 will be compared to ground controls to ascertain how the presence of V. fischeri alters the host transcriptional and metabolic response to spaceflight conditions during early development. Significance of Project - The proposed work has several implications for the field of space biology and directly relates to NASA Space Biology Science Plan and Decadal Survey, specifically the Microbiology (MB) and Cell Molecular Biology (CMB) program element. First, by examining how the normal bacteria-induced development of animals is mediated under spaceflight conditions, we can begin to determine the sensitivity of these associations to perturbations in the space environment. Additionally, by removing gravity as a constant we can address the question of whether gravity obscures aspects of bacteria-induced animal development that may otherwise go undetected under terrestrial conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work has several implications for the field of space biology and life on Earth. First, by examining how healthy animal-microbe interactions are initiated and maintained under microgravity conditions, we can begin to determine the sensitivity of these associations to perturbations in the space environment. There is a gap in our understanding of the effects that the space environment has on the microbiome of animals. As there are more than 2000 identified bacterial species that form associations with humans, there are fewer than 100 species identified as human pathogens, yet most microgravity studies have focused on pathogens. This leaves a large array of undescribed commensal organisms that form interactions with animal cells, which may, or may not, be impacted by microgravity conditions. Here in this proposal, we ask the question of how does microgravity alter key bacteria-induced developmental events within the host animal? In other words, what happens to mutualistic associations when gravity is removed? Secondly, by removing gravity as a constant we can address the question of whether gravity obscures aspects or cues of bacteria-induced animal development that may otherwise go undetected under terrestrial conditions. These new insights from simplified animal model systems can then be applied to our understanding of human physiology in the space environment.
	Science Objective 1. Characterize the changes in the transcriptome of the host squid in the presence and absence of symbiotic bacteria exposure under microgravity conditions. a. Transcriptomes are relatively stable during spaceflight. A PCA plot was generated of all 32 of the RNASeq libraries generated as part of this project (Fig. 1A). The PCA plot shows that the libraries cluster into two distinctive populations one for flight and one for ground. Interestingly, only 14% of the variation in the RNASeq libraries can be explained by the spaceflight. These results suggest most transcripts within the transcriptome are, in general, relatively stable, and not differentially expressed during the spaceflight launch and transition to the microgravity environment. b. Differential gene expression associated with neurogenesis is being expressed earlier under spaceflight conditions in the presence of symbiotic bacteria compared to gravity controls. The presence of the symbiotic bacteria triggered differential gene expression of several pathways associated with neurogenesis including Axon Guidance, Neuron Differentiation and Axonogenesis in the presence of the symbiont compared to animals that were maintained aposymbiotic conditions. c. Aposymbiotic animals have a higher rate of differential gene expression than symbiotic animals in flight compared to ground controls. A comparison of symbiotic and aposymbiotic transcriptomes revealed that at all time points tested, except 0 h, there was approximately 3 x the number of differentially expressed genes in aposymbiotic animals. For example, at 12, hours there were n= 29 significantly differentially expressed transcripts in apo and n = 10 in sym at 12 h. Significance was defined as FDR >0.05 and Fold Change > 2. This result corresponds to several ground-based studies completed by the PI in previous research. The results suggest the symbiont could be playing a role in regulating gene expression in the host animal during the stress of spaceflight. d. Pronounced increase in genes associated with stress response, lipid processing and immune function during spaceflight in aposymbiotic animals. Of the genes that were differentially expressed in the flight conditions, there was an

Task Progress:	enrichment of pathways associated with oxidative stress responses in the aposymbiotic animals compared to those flight animals exposed to the natural symbiont (Fig. 3). Additionally, there was a pronounced increase in the expression of an apolipophorin-encoding gene in aposymbiotic animals during flight. Apolipophorin is a Pattern Recognition Receptor immune protein in animals that binds lipopolysaccharide. It's often associated with the host immune system but is often increased under physical stress conditions. The precise role of apolipophorin increase in the host animal during spaceflight is not yet know. Follow up research over the next year will attempt to identify the specific function of this protein during spaceflight. Interestingly, these stress responses were not present in symbiotic light organs, again suggesting the presents of mutualistic bacteria may be mitigating the stress effects of spaceflight. Science Objective 2. Ascertain changes in the metabolome of host squid light organ in the presence and absence of symbiotic bacteria under spaceflight conditions. A. Research Progress and Analyses - Using the same flight samples as described in Objective 1, an untargeted metabolomics approach was used to reveal differential metabolic activity in symbiotic and aposymbiotic light organs. A total of four light organs were used for each time point. A new dual analysis was successfully completed by the University of Florida Southeast Center for Integrated Metabolomics in which the samples were split and an untargeted metabolomic and lipidomic analysis was performed on each replicate sample. 1. Metabolomic and lipidomic differences observed between ground and flight samples. Preliminary analysis of the samples revealed that the largest difference between the samples was the environmental condition of spaceflight. Two distinct populations were visible in both analyses. Most of the recovered metabolites expressed in the host squid during spaceflight were novel. Efforts are currently underway to examine those specific metabolites and lipids that were significantly different between time points and treatments. 2. Lipidomics analyses reveal a pronounced shifts in ceramides during spaceflight Analysis of the lipidomics and metabolomics data sets have been challenging as most of the differentially expressed small molecules are unknown and are not able to be identified. Work is ongoing to identify the specific metabolites against a database of known standards and as we generate hypotheses about the unknown molecules, we are working with the UF core facility to run the standards to identify the unknown molecules. Despite these challenges, we have identified an important group of lipids, the ceramides, that have a very pronounced increase in abundance in the symbiotic animals during spaceflight. Ceramides are lipid molecules that have a very diverse range of functions in cells as they can be critical molecules for cell differentiation, apoptosis, and neurogenesis. The fact that the increase in ceramide production is coinciding with the increase expression of gene expression in cell membrane and neurogenesis suggest the ceramides might be playing an important role in neuron development in the host animal and that is increase under spaceflight conditions in the presence of symbiotic microbes. Science Objective 3. Build integrative networks of transcriptomes and metabolomes under different flight and ground-based treatments to elucidate crucial pathways that differentially respond to the space flight environment. The networks generated from the results of Objectives 1 and 2 will be compared to ground controls to ascertain how the presence of V. fischeri alters the host transcriptional and metabolic response to spaceflight conditions during early development. At this point we are still in the process of analyzing the data derived from the RNASeq, metabolomics, and lipidomics data and have not yet been able to integrate all data into a network. Due to the large amounts of data analysis required to separately examined the transcriptomics, metabolomics and lipidomics we have not sufficiently integrated all the data but will continue to do so over the next year, even though the grant has ended.

Bibliography Type:	Description: (Last Updated: 04/20/2026)
Articles in Peer-reviewed Journals	Bongrand C, Foster JS. "Modeled microgravity impacts Vibrio fischeri population structure in a mutualistic association with an animal host." Environ Microbiol. 2023 Dec;25(12):3269-79. https://doi.org/10.1111/1462-2920.16522 ; PMID: 37828645 , Dec-2023
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